# Thermodynamics & Statistical Mechanics

## Four Laws of Thermodynamics

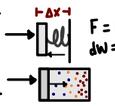
$T$ - temperature
if $A \leftrightarrow B$, $B \leftrightarrow C$ then $A \leftrightarrow C$

energy, heat flow
$\left[U = \int D(\varepsilon) N(E) \cdot E \, dE\right]$

$\mu$ - chemical potential
number flow (atoms, particles)
$\left[N = \int D(\varepsilon) N(E) \, dE\right]$

⓪ equilibrium (implies irreversibility)
asymmetry in time

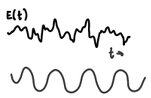
① energy conservation $dW = dQ$
$Q$ = internal energy, $\Delta$work = $\Delta$heat
→ randomized - 'heat'
total change $U$
= sum of work done on system + heat flow




② entropy (always stays the same or increases over t)

$\Delta S \geq 0$ $\quad S = k_B \log_e \Omega$ ← # of equivalent states
↳ $S$ is a measure of 'degree of disorder'

$\log \Omega$ counts # of 'bits' needed to describe the states

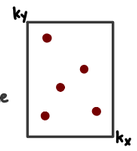



→ $S = -k_B \log \frac{1}{\Omega}$ → probability of being in any specific unique state
* what if states are not equally probable? → weighted average

- $S = \sum_i P_i S_i = -k_B \sum_i P_i \log P_i \Rightarrow S = -k_B \sum_i N_i \log N_i$
  'Von Neumann Entropy' $\quad$ QM "Diagonal Entropy"

$\Delta S \geq 0$ - change in entropy always positive
$\frac{dS}{dt} \geq 0$ - entropy always increases
} Proof: Boltzmann's H-Theorem

③ Absolute Zero, Absolute Limit
$S = 0$ @ $T = 0$
$\log(\Omega = 1) = 0$




QM:
$\psi(x)$ - complex $|\psi| e^{i\varphi}$
$N(x) \propto |\psi(x)|^2$
$\psi(x_1)\psi(x_2)$
} both methods lose system information

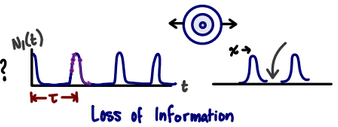
Classical: velocity of individual particles unknown - course grained

Poincare' Cycles (?) - appears to disagree with Boltzmann's H-Theory

① Poincare' time is long for even simple systems

② Poincare' time → ∞ in ∞-large systems

③ where did Boltzmann's equations go wrong?




one can prove (both classically and w/ QM)
in any finite system that has time-reversible dynamics: all behaviors repeat in cycles

# Boltzmann's H-Theorem

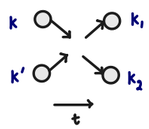
'proof of 2nd Law' → same tools as detailed balance

$S = -k_B \sum_k N_k \log N_k$ - 'classical limit' of QM Entropy

2nd Law: $\frac{\partial S}{\partial t} \geq 0 \Rightarrow \frac{\partial S}{\partial t} = -k_B \sum_k \left( \frac{\partial N_k}{\partial t} \ln N_k + \underbrace{N_k \frac{\partial}{\partial t} \ln N_k}_{N_k \frac{1}{N_k} \frac{\partial N_k}{\partial t}} \right) \Rightarrow \frac{\partial}{\partial t}\left(\sum_k N_k\right)$
→ = 0 if N is conserved

recall, rate of change:

$$\frac{\partial N_k}{\partial t} \propto \sum_{k', k_1, k_2} \left[ \underbrace{-N_k N_{k'} (1 \pm N_{k_1})(1 \pm N_{k_2})}_{\text{outflow}} + \underbrace{N_{k_1} N_{k_2} (1 \pm N_{k'})(1 \pm N_{k'})}_{\text{inflow}} \right]$$




$$\frac{\partial S}{\partial t} \propto \sum_{k, k', k_1, k_2} \left[ N_{k_1} N_{k_2} - N_k N_{k'} \right] \ln N_k$$

→ terms including a particular set of $k, k', k_1, k_2$

$$\underbrace{(\ln N_k - \ln N_{k'} - \ln N_{k_1} - \ln N_{k_2})}(N_{k_1} N_{k_2} - N_k N_{k'})$$

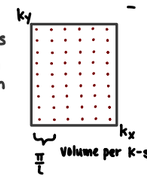
$$\ln\left(\frac{N_k N_{k'}}{N_{k_1} N_{k_2}}\right) \underbrace{(N_{k_1} N_{k_2} - N_k N_{k'})}$$

if this is larger → term is positive, then log must be negative!

} cumulative term / relation is ALWAYS Negative!
remember: $\frac{\partial S}{\partial t} \geq 0$

## Ideal Classical Gas

- for a given $T$, what is average energy per particle?

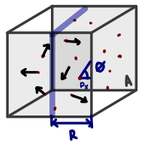



volume per k-space is $\left(\frac{2\pi}{V}\right)^3$

$$N = \frac{V}{(2\pi)^3} \int d^3k \, N(E_k)$$

↑ avg. value $N(E) = e^{-(E_k - \mu)/k_B T} = (e^{\mu/k_B T}) e^{-E_k/k_B T}$

## Ideal Gas Law - relation of pressure, T, and A




$P = \frac{F}{A} = \frac{1}{A} \frac{dp}{dA}$, atom has: $p_x, p_y, p_z$
• $p_x$ contributes to $F_x$

density $\frac{1}{2} \frac{N}{V} \lambda$

N particles hitting wall per unit time $\quad \lambda = v_x \Delta t$

Total Momentum per Unit Time

$$\frac{F}{A} = \frac{1}{2} \frac{N}{V} v_x \Delta t \frac{(2mv_x)}{\Delta t}$$

$$P = \frac{N}{V} m v_x^2 = \frac{N}{V} k_B T$$

$PV = Nk_BT = nRT$ $\quad$ * $R = N_A k_B$

* equipartition:
$E = \frac{1}{2} m\bar{v}_x^2 + \frac{1}{2} m\bar{v}_y^2 + \frac{1}{2} m\bar{v}_z^2 = \frac{3}{2} k_B T$
isotropic system:
$\bar{v}_x^2 = \bar{v}_y^2 = \bar{v}_z^2 \rightarrow 3(mv_x^2) = 3k_BT$

① Change of Variables $k \rightarrow \varepsilon$:

$d^3k = 4\pi k^2 dk$ for isotropic average over angle
↳ direction doesn't matter

$\Rightarrow (d^3k = k^2 dk \sin\theta d\theta d\varphi)$: $E_k = \frac{\hbar^2 k^2}{2m} \rightarrow \frac{dE_k}{dk} = \frac{\hbar^2 k}{m} \rightarrow dE_k = \frac{\hbar^2 k}{m} dk$

$dk = \frac{m}{\hbar^2 k} dE$

$- k = \frac{\sqrt{2mE_k}}{\hbar}$

- $N = \frac{V}{(2\pi)^3} 4\pi \int k^2 dk \, e^{\mu/k_BT} e^{-E_k/k_BT}$

$= \frac{V}{(2\pi)^2} 4\pi \int k^{\cancel{2}} \frac{m}{\hbar k} dE_k \, e^{\mu/k_BT} e^{-E_k/k_BT}$

$= \frac{V}{(2\pi)^2} e^{\mu/k_BT} \int_0^\infty \frac{\sqrt{2mE_k}}{\hbar} \frac{m}{\hbar} dE_k \, e^{-E_k/k_BT} \Rightarrow \frac{V}{(2\pi)^2} e^{\mu/k_BT} \int_0^\infty \frac{\sqrt{2mk_BT}}{\hbar} m \sqrt{\varepsilon} \, e^{-\varepsilon} (k_BT) d\varepsilon$

$$N = \frac{V}{(2\pi)^3} \int d^3k \, e^{-(\varepsilon - \mu)/k_BT}$$

$$= \frac{V}{(2\pi)^3} \cdot e^{\mu/k_BT} \int_0^\infty dE \frac{\sqrt{2mE}}{\hbar^3} \cdot e^{-\varepsilon/k_BT}$$

$$N = e^{\mu/k_BT} \int_0^\infty \underbrace{D(E)}_{\text{density of states}} dE \, e^{-\varepsilon/k_BT}$$

→ units of $\frac{1}{E}$
(# of states falling between $(E, E+dE)$) $\quad \frac{\hbar^2 k^2}{2m}$

$$\Rightarrow N = \frac{V}{(2\pi)^2} e^{\mu/k_BT} \frac{\sqrt{2} m^{3/2} (k_BT)^{3/2}}{\hbar^3} \left[\int_0^\infty \varepsilon^{1/2} e^{-\varepsilon} d\varepsilon\right] = \frac{\sqrt{\pi}}{2}$$

$$= \frac{V}{(2\pi)^2} e^{\mu/k_BT} \frac{\sqrt{2} m^{3/2} (k_BT)^{3/2}}{\hbar^3} \left[\int_0^\infty \varepsilon^{1/2} e^{-\varepsilon} d\varepsilon\right] = \frac{\sqrt{\pi}}{2}$$

* $\varepsilon = \frac{E}{k_BT}$
→ $E = \varepsilon k_BT$

- $U = \frac{V}{(2\pi)^2} \int d^3k \, e^{-(E-\mu)/k_BT} E_k$

$= \frac{V}{(2\pi)^2} e^{\mu/k_BT} \int dE \frac{\sqrt{2mE}}{\hbar^3} E \, e^{-E/k_BT}$

$= \frac{1}{2\pi^2} e^{\mu/k_BT} \frac{\sqrt{2} (mk_BT)^{3/2}}{\hbar^3} \int_0^\infty \varepsilon^{1/2} (\varepsilon k_BT) e^{-\varepsilon} d\varepsilon$

$k_BT \int_0^\infty \varepsilon^{3/2} e^{-\varepsilon} \rightarrow \frac{3\sqrt{\pi}}{4}$

average energy / particle $= k_BT \frac{\sqrt{\pi}/2}{3\sqrt{\pi}/4}$

## Connection between Entropy (S) and Heat (Q)

- $S = -k_B \sum_k N_k \log N_k$ $\quad$ * $N_k = e^{\mu/k_BT} e^{-E/k_BT}$

$\Rightarrow S = -k_B \sum_k e^{-(E-\mu)/k_BT} \left[\frac{(-E+\mu)}{k_BT}\right]$

$= \frac{-\mu k_BT}{k_BT} N + \frac{k_B}{k_BT} U$

$\Rightarrow TS = -\mu N + U$

↳ $\underbrace{\mu N}_{F \text{ - Free energy}} = U - TS$

$\Rightarrow dS\big|_N = \frac{dU}{T}$

→ $dU = TdS$

* $\Delta S \geq 0$
→ $\Delta F \leq 0$ @ constant $U$
equilibrium ⟶ minimizing F

ordered states:
(very few equilibrium states)
- low S
- high F

## Heat Capacity & Heat Flow

↳ $C_V = \frac{dQ}{dT}\Big|_V$ $\quad$ high heat capacity: small $\Delta T$ for large $\Delta Q$

work done is zero: $dW = PdV = \frac{F}{A}(Adx) = Fdx$

energy outflow from system $\quad$ Pressure

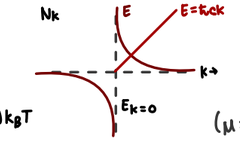
↳ $C_V = \frac{dU}{dT}$ and in an ideal gas: $U = \frac{3}{2} N k_BT$

$C_V = \frac{3}{2} N k_B$

## Heat Capacity of Photons and Phonons

recall, $U = \frac{V}{(2\pi)^3} \int d^3k \, N_k E_k$

where $E_k = \hbar ck$ for photons, not: $E_k = \frac{\hbar^2 k^2}{2m} = \frac{1}{2} mv^2 = \frac{p^2}{2m}$




can't use $N_k \simeq e^{-(E-\mu)/k_BT}$

planck: $N_k = \frac{1}{e^{E_k/k_BT}}$

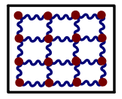
($\mu = 0$ for system with N not conserved)

$\Rightarrow U = \frac{V}{(2\pi)^3} (4\pi) \int k^2 dk \cdot N(E_k) E_k$

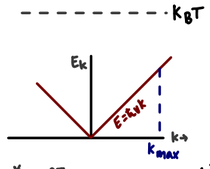
$\propto \frac{V}{2\pi^2} \int E^3 dE \frac{1}{e^{E/k_BT} - 1}$ $\quad$ change $\varepsilon = E/k_BT$, $d\varepsilon = dE/k_BT$

$\rightarrow U \propto (k_BT)^4 \int_0^\infty \varepsilon^3 d\varepsilon \frac{1}{e^\varepsilon - 1}$ $\quad E = \varepsilon k_BT$

$U \propto (k_BT)^4$ 'Stephan-Boltzmann Law' → $C_V = \frac{dU}{dT}\Big|_V \propto T^3$



λ - wavelength




* $k = \frac{2\pi}{\lambda}$, $k_{max} = \frac{2\pi}{\lambda_{max}}$

$e^{i(0)}$ $\quad e^{i\pi}$

→ $U = \frac{V}{2\pi^2} \int_0^{k_{max}} k^2 dk \, N_k E_k$

* are photons cutoff in k?
'planck distance'

low T limit → $C_V \propto T^3$:

$\Rightarrow U = \frac{V}{2\pi^2} \int_0^{k_{max}} k^2 dk \frac{k_BT}{E_k} E_k = \frac{V}{2\pi^2} (k_BT) \int_0^{k_{max}} k^2 dk \simeq \frac{4\pi}{3} k_{max}^3$

high T limit:

$N_k = \frac{1}{e^{E_k/kT} - 1} = \frac{1}{\left(1 + \frac{E_k}{k_BT} + \ldots\right) - 1} \simeq \frac{k_BT}{E_k}$

↑ Taylor Series for exp. f(x)